

Hybrid RAG

Overall Mean Score: 0.4633 — #1 Ranked

- Combines BM25 keyword search (top-10) and ChromaDB semantic search (top-10), fuses both lists using Reciprocal Rank Fusion (RRF, k=60) and passes the top-5 documents to the LLM
- Achieved the highest Context Precision (0.5445) and Context Recall (0.5637) across all pipelines, indicating the retrieved documents were both relevant and complete
- Faithfulness score of 0.4072 is significantly higher than all other pipelines, meaning generated answers are better grounded in the retrieved context
- Answer Relevancy of 0.5897 — highest among all pipelines — reflects that responses stay on-topic and address what was asked
- Factual Correctness of 0.1563 is highest overall but still low, driven by ROUGE-L mismatch between verbose LLM answers and short ground-truth answers
- + Best retrieval pipeline for mixed query types — BM25 handles exact keyword matches (order IDs, category names, seller IDs) while semantic search handles paraphrased or conceptual queries
- + RRF fusion boosts documents that rank highly in both BM25 and semantic lists, reducing noise from either retrieval channel alone
- + Single LLM call per query — lower cost than HyDE or MultiQuery despite stronger retrieval
- + BM25 index is fully offline and fast; no GPU needed for keyword retrieval
- + Consistently outperforms all other pipelines on easy, medium, and hard difficulty questions
- ✗ Requires maintaining two separate indexes (BM25 pickle + ChromaDB vector store), increasing storage and setup complexity
- ✗ BM25 tokenisation is language-sensitive; performance may degrade on queries with heavy domain jargon not present verbatim in documents
- ✗ Hallucination score (0.5928) is the lowest across pipelines, meaning it produces fewer hedging or uncertain statements — good for confidence, but worth monitoring
- ✗ RRF k=60 is a fixed hyperparameter; suboptimal for very short or very long candidate lists without tuning

Reranking RAG

Overall Mean Score: 0.3655 — #2 Ranked

- Retrieves an initial pool of 20 candidate documents from ChromaDB, then re-scores all 20 using a cross-encoder (cross-encoder/ms-marco-MiniLM-L-6-v2) and selects the top-5 for generation
- Context Precision of 0.4095 — second highest after Hybrid RAG — reflects the cross-encoder's ability to correctly promote the most relevant document to position 1
- Faithfulness (0.2432) and Context Recall (0.3565) are slightly better than Naive and HyDE, suggesting reranking marginally improves grounding
- Performs best on factual easy questions where a single document clearly contains the answer; the cross-encoder confidently promotes it to rank 1
- On analytical and comparison questions, reranking provides less benefit because multiple supporting documents are needed, not just the single best one
- + Cross-encoder reranking captures fine-grained relevance that bi-encoder embeddings miss, especially for nuanced or multi-clause queries

- ✚ No additional LLM call required beyond the final answer generation — the cross-encoder is a local model with low latency
- ✚ Straightforward pipeline extension on top of any vector store; reranker is plug-and-play
- ✚ Provides explainability through reranking scores — can inspect why each document was promoted or demoted
- ✗ Cross-encoder must score all 20 candidates at inference time, adding latency proportional to pool size
- ✗ If the correct document is not in the initial top-20 semantic recall pool, reranking cannot recover it — limited by upstream recall
- ✗ Context Recall (0.3565) is not much better than Naive RAG (0.3338), meaning the initial semantic retrieval is still the bottleneck
- ✗ Cross-encoder model (ms-marco-MiniLM-L-6-v2) was trained on MS MARCO web data; may not generalise perfectly to structured e-commerce domain documents

Naive RAG

Overall Mean Score: 0.3486 — #3 Ranked

- Single-stage retrieval using ChromaDB cosine-similarity search with all-MiniLM-L6-v2 embeddings; no reranking, no query expansion, one LLM call per query
 - Context Precision of 0.3225 and Context Recall of 0.3338 are the lowest retrieval scores — correct documents are retrieved but often ranked below noisy ones
 - Faithfulness of 0.2679 and Answer Relevancy of 0.5077 establish the baseline that all other pipelines are measured against
 - Performs well on easy factual queries where the query string closely matches the document text (high embedding similarity to ground-truth document)
 - Hallucination score of 0.7321 is high — the model frequently generates answers not directly supported by retrieved context, especially on medium and hard questions
 - ✚ Simplest architecture: one ChromaDB collection + one embedding model + one LLM call — minimal dependencies and fast to deploy
 - ✚ Lowest operational cost: no extra models, no BM25 index, no reranking step
 - ✚ Shared embedding model between indexing and querying eliminates representation mismatch
 - ✚ Strong baseline — despite simplicity, matches HyDE and MultiQuery on overall score
 - ✚ Easy to debug: clear one-to-one mapping between query embedding and retrieved document
 - ✗ Pure semantic search fails on exact-match queries (order IDs, seller IDs, specific numeric values) where BM25 would excel
 - ✗ No mechanism to recover from a poor initial retrieval; the top-5 results are passed directly to the LLM regardless of quality
 - ✗ Embedding similarity treats all tokens equally — stop words and irrelevant terms can dilute similarity for domain-specific short queries
 - ✗ Lowest Context Precision among all pipelines; noisy documents frequently occupy top-k slots
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HyDE RAG

Overall Mean Score: 0.3466 — #4 Ranked

- Generates a hypothetical answer document from the query using the LLM (call 1), embeds that hypothetical document, and uses its embedding to retrieve real documents from ChromaDB (call 2 for final answer) — 2 LLM calls total
- Theory: a hypothetical answer embedding is closer in vector space to actual answer documents than the raw question embedding
- Context Recall of 0.3437 is marginally better than Naive RAG (0.3338), suggesting the hypothetical document occasionally helps surface additional supporting documents
- Performs comparably to Naive RAG on most metrics despite twice the LLM cost, indicating the hypothetical document does not provide a substantial retrieval advantage on this structured e-commerce dataset
- Hallucination score of 0.7659 is the highest across all pipelines — the model most frequently generates answers not grounded in retrieved documents, likely because hypothetical docs introduce factual drift in retrieval
- + Effective when the original query is vague, abstract, or uses different vocabulary than the knowledge base documents
- + Hypothetical document generation can act as a zero-shot query expansion without explicit expansion logic
- + Works purely with the same vector store infrastructure as Naive RAG — no additional index or model required
- + Well-suited to open-ended analytical questions where framing the answer space helps narrow retrieval
- ✗ Doubles the number of LLM API calls, doubling token cost and latency
- ✗ Hypothetical documents can hallucinate facts that actively misdirect retrieval toward wrong documents — especially harmful on precise factual queries
- ✗ On a structured, fact-dense e-commerce knowledge base, query terms like order IDs and seller hashes are poor inputs for generating useful hypothetical answers
- ✗ Faithfulness (0.2341) is the lowest of all pipelines — hypothetical doc drift compounds into lower answer grounding
- ✗ No improvement over Naive RAG on Context Precision (0.3117 vs 0.3225) despite higher computational cost

MultiQuery RAG

Overall Mean Score: 0.3457 — #5 Ranked

- Expands each user query into 4 semantically diverse sub-queries using the LLM (call 1), retrieves top-10 documents per sub-query from ChromaDB, fuses all results with RRF (k=60), and selects the top-5 for generation (call 2)
- Lowest overall mean score (0.3457) despite using query expansion and RRF fusion — additional sub-queries on this dataset introduce noise rather than improve recall
- Context Precision of 0.3132 is the lowest after HyDE — expanding to 4 queries increases the chance of retrieving off-topic documents that dilute the fused result
- Context Recall (0.3341) is nearly identical to Naive RAG (0.3338) — the query expansion adds complexity without meaningfully improving coverage of the ground-truth document

- Hallucination score of 0.7545 is high, second only to HyDE, indicating generated answers are frequently not well-grounded in the retrieved multi-query context
- ✚ Multi-angle retrieval is beneficial when queries are ambiguous or when the knowledge base uses varied vocabulary for the same concept
- ✚ RRF fusion across 4 sub-query result sets increases the chance that the correct document appears in at least one retrieval list
- ✚ Query expansion is entirely LLM-driven — no additional index, model weights, or infrastructure beyond the base vector store
- ✚ Particularly effective for complex multi-part analytical questions where different sub-questions target different aspects of the answer
- ✗ Two LLM calls per query (expansion + generation) doubles cost and latency with no score improvement over Naive RAG on this dataset
- ✗ LLM-generated sub-queries may paraphrase the original but not actually diversify the retrieval angle, especially for short factual queries like 'What is the total payment for order X?'
- ✗ Largest retrieval footprint (up to 40 candidate documents across 4 sub-queries before RRF) — most computationally expensive retrieval step
- ✗ On structured e-commerce data with exact-match needs (IDs, numeric values), query expansion generates semantically similar but lexically different queries that all fail to match the ground-truth document